

Encircled energy in mophongo

Mophongo technical note

2026 August 10

Abstract

Three arrays in the template-fitting chain have finite support: the two PSF stamps and the source template. Let f be the total flux of a source and A the amplitude returned by the fit, the coefficient multiplying its unit-sum template. For a point source $A = f S_{\text{lo}}$, where S_{lo} is the fraction of the low-resolution PSF enclosed by its stamp. The total flux is therefore $f = A/S_{\text{lo}}$: the amplitude is divided by S_{lo} , not multiplied. This holds exactly, for any weight map and independently of how the high-resolution stamp is truncated, so one scalar converts amplitude to total flux. It is measured on the drizzled stamp at each source position and applied once. A second quantity, the fraction of the normalised template that survives blanking of neighbouring segments, is recorded but deliberately not applied: the amplitude does not scale with it. We give the measurements behind each choice and show that the residual error is set by the matching kernel rather than by any encircled-energy term: a Fourier-window kernel biases the flux scale by 2.3 to 3.3%, a regularised Wiener kernel by 0.16%, flat in stamp size.

1 The correction

Write Φ_{hi} and Φ_{lo} for the true PSFs of the detection (high-resolution) and measurement (low-resolution) bands, each integrating to unity over an infinite aperture, and L for the side of the square stamp set by `psf_size`. The stamps $P_{\text{hi}} = \Phi_{\text{hi}}|_L$ and $P_{\text{lo}} = \Phi_{\text{lo}}|_L$ hold

$$S_{\text{hi}} = \int_L \Phi_{\text{hi}}, \quad S_{\text{lo}} = \int_L \Phi_{\text{lo}}, \quad (1)$$

both below one: they are the fractions of each PSF that the stamp encloses, 0.95 and 0.85 for JWST F444W/F770W at $L = 1.5''$, rising to 1.00 and 1.00 at $8''$ (Figure 1). The pipeline normalises the stamps to unit sum, $p_{\text{hi}} = P_{\text{hi}}/S_{\text{hi}}$ and $p_{\text{lo}} = P_{\text{lo}}/S_{\text{lo}}$, and builds the matching kernel k from those shapes, so that $p_{\text{hi}} \otimes k \simeq p_{\text{lo}}$, then renormalises it to $\sum k = 1$.

The template T is the source model in the detection band, normalised to unit sum at extraction. For a point source whose template is extended over the stamp, $T = p_{\text{hi}}$, and the model fitted to the low-resolution image is

$$M = T \otimes k = p_{\text{lo}} = \Phi_{\text{lo}}|_L / S_{\text{lo}}, \quad (2)$$

a unit-sum array proportional to the truth inside L and zero outside. The fit solves for the amplitude A that scales M onto the data. With f the source's total flux, W the inverse-variance weight map and $\langle a, b \rangle = \sum_{\text{pix}} a b$,

$$A = f \frac{\langle M, W \Phi_{\text{lo}} \rangle}{\langle M, W M \rangle} = f S_{\text{lo}}, \quad (3)$$

exactly, and for any W , because $M \propto \Phi_{\text{lo}}|_L$ and M vanishes wherever the two differ. The amplitude is low by exactly the fraction of the PSF that the stamp omits, so $f = A/S_{\text{lo}}$. The $1/S_{\text{hi}}$ introduced in forming p_{hi} is replaced by $1/S_{\text{lo}}$ by the kernel, whose target is the low-resolution shape; this is why only the low-resolution stamp appears.

This is a property of the unit-kernel convention. With a native-sum kernel, $\sum k = S_{lo}/S_{hi}$, the same algebra gives $A = f S_{hi}$ and it is S_{lo} that cancels. The two differ by S_{hi}/S_{lo} , which is 4.6% on the UDS products, so the convention must be stated wherever the divisor is applied.

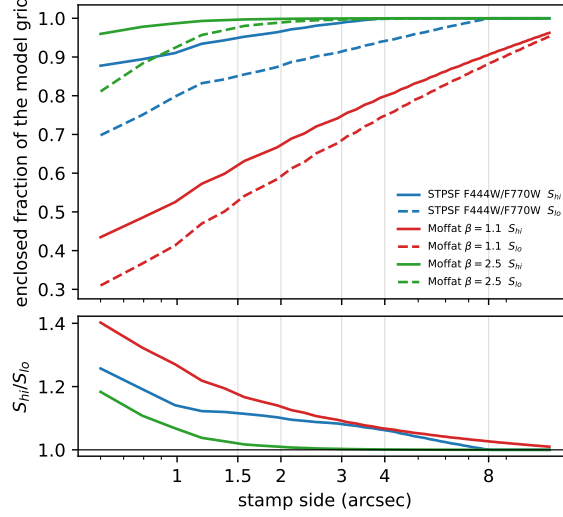


Figure 1: Top: the enclosed fraction of each PSF within a square stamp, S_{hi} solid and S_{lo} dashed, against the stamp side L . Bottom: their ratio. The low-resolution stamp always encloses less, and the two converge only for stamps several times the PSF width. Vertical lines mark the stamp sizes tested in Section 3. The F444W curve saturates at $4.089''$, the field of view of its model grid.

2 What the pipeline does

The absolute scale is measured once, carried alongside the arrays, and divided out once. Nothing in between is absolutely calibrated.

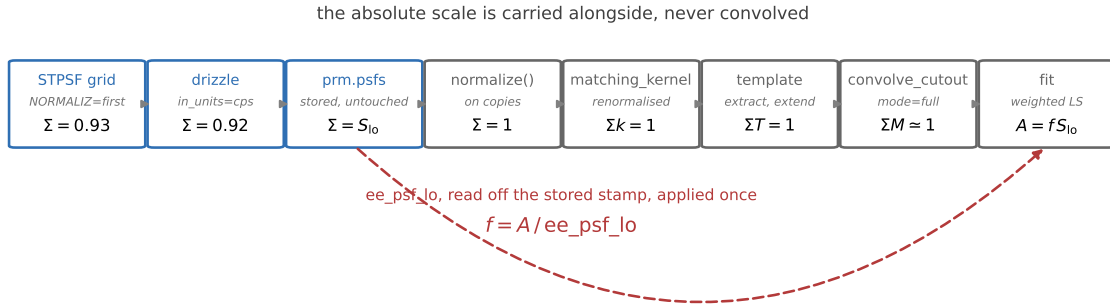


Figure 2: Blue stages hold absolute flux, grey stages hold shapes. The stored PSF maps are not renormalised; only the copies passed to `matching_kernel` are. The dashed path is the absolute scale, which bypasses the convolution and re-enters once as `ee_psf_lo`.

Unit-sum inputs to the kernel, native sums on disk. Normalising the inputs rescales the kernel and nothing else: the ratio of the two DC sums equals S_{hi}/S_{lo} to all printed digits and the shapes agree to below 10^{-15} , since the regularisation parameter is scaled by $\max |H|^2$ and is therefore insensitive to the input scale. Native-sum inputs instead put S_{hi}/S_{lo} into $\sum k$, where

stage	array	sum	code
PSF model	STPSF grid	0.93 absolute	<code>data/PSF/*GRID*.fits</code>
drizzle	stamp	S_{lo} absolute	<code>DrizzlePSF, in_units="cps"</code>
store	<code>prm.psfs</code>	unchanged	<code>build_psfs</code>
measure	<code>ee_box, ee_rlim</code>	per region	<code>PSFRegionMap.refresh_ee</code>
match	<code>normalize()</code> copies	1	<code>build_kernels</code>
kernel	k	1, renormalised	<code>matching_kernel</code>
template	T	$ee_tmpl \leq 1$	<code>extract_templates</code>
convolve	$M = T \otimes k$	$\simeq 1$	<code>convolve_cutout</code>
record	<code>ee_psf_lo, ee_tmpl</code>	per template	<code>convolve_templates</code>
solve	A	$f S_{lo}$	<code>SceneFitter</code>
report	A/ee_psf_lo	f	<code>Pipeline.run</code>

Table 1: The chain in code terms.

it cannot be distinguished from a regularisation error, and they place the amplitude on the S_{hi} convention while the divisor is S_{lo} .

The kernel is renormalised to unit sum. With unit-sum inputs $\sum k$ already lands within a part in 10^3 of one; dividing it out removes the residual regularisation DC so that the kernel carries no flux scale of its own, leaving `ee_psf_lo` as the only factor between amplitude and total flux. It is free: the fitted amplitude is invariant under a rescaling of the model, and every recovered flux in Section 3 is unchanged to five decimals by it. It also says nothing about whether the kernel has the right *shape*. A window kernel has $\sum k = 1$ and still biases the flux scale by 2.2% (Section 4); unit sum is a necessary condition, not a sufficient one.

Encircled energy is measured on the stamp in use. The STPSF grids carry `NORMALIZ = first`, so $\sum(\text{grid})/\text{OVERSAMP}^2$ is absolute, and drizzling with `in_units="cps"` preserves it. Summing the stored stamp gives the absolute value with no separate calibration step, and it includes the edge taper, the negative clipping, the drizzle kernel and the Fourier blur. A tabulated value would describe a different array from the one being convolved.

Stored per region, recorded per template. S_{lo} varies over the field: across the 294 regions of the UDS F770W map `ee_box` spans 0.90818 to 0.92235 about a median of 0.91710, and across the 1694 F444W regions 0.94722 to 0.96598. `PSFRegionMap.refresh_ee` measures every stamp when `psfs` is set, so `get_ee_box(ra, dec)` costs a spatial query and one array index, 11.8 μs against 11.3 μs for `get_psf` at the same call site. `convolve_templates` writes `ee_psf_lo` onto each convolved template, which is the per-source, per-band object; the detection template is neither. `flux.<i>total` then divides by that value rather than by a filter-level mean, which would mis-scale sources by -1.0 to $+0.6\%$ across the UDS field. `ee_psf_lo` defaults to `nan`, so a template that never saw a PSF map falls back to the filter mean and is counted in the run log rather than silently going uncorrected.

ee_box and ee_rlim. `ee_box` is the sum over the square stamp and divides the fitted amplitude, since the model occupies the square. `ee_rlim` is the sum inside the inscribed circle and is the matching quantity for a circular aperture. On region 0 of the UDS F770W map they are 0.91826 and 0.91060, the difference being the corner flux (Figure 3).

ee_tmpl is a diagnostic, not a divisor. Templates are normalised over the whole stamp and then blanked on neighbouring segments, so that wing flux landing on a neighbour is not fitted twice. What survives is $ee_tmpl = \sum T \leq 1$, recorded per template beside `ee_psf_lo`. It is tempting to write $f = A/(ee_psf_lo \cdot ee_tmpl)$, but the amplitude does not scale with the

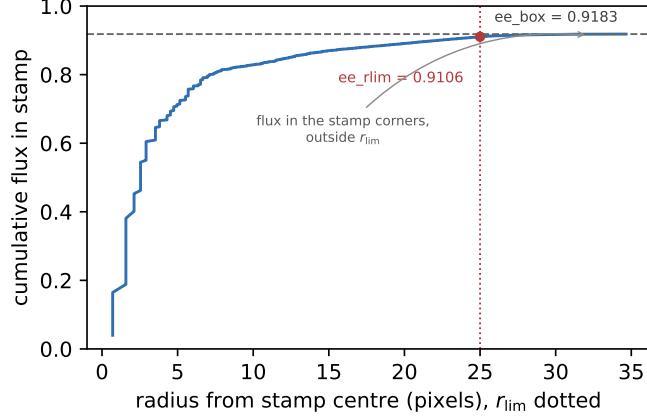


Figure 3: Cumulative flux of a drizzled F770W stamp against radius. The curve is flat well inside the stamp edge, so both measures are insensitive to small changes in `psf_size`.

blanked flux. It scales with the weighted leverage of the blanked pixels, and neighbour segments lie in the faint wings, where the flux is a few percent and the contribution to χ^2 is negligible; least squares recovers almost all of it by scaling the retained part up. Measured on an injected point source with a neighbour segment blanked 12 pixels out, `ee_tmpl` = 0.9928 while A/S_{lo} = 1.0021, so dividing by `ee_tmpl` would move a 0.2% error to 0.9%. Solving a blended pair jointly, with each template blanked on the other's segment, gives A/S_{lo} = 1.010 at 12 pixels separation and 1.033 at 8 pixels, against 1.017 and 1.067 after dividing by `ee_tmpl`. The correction over-corrects in every configuration tested. The residual at small separations is blend cross-talk, which no per-source scalar removes; `ee_tmpl` is the right quantity to flag it with, and $f = A/\text{ee_psf_lo}$ remains the correction.

Kernel method. `build_kernels` uses `method="wiener"` with the regularisation scanned once over the median of the hi and lo PSF stacks and reused for every region, since a per-region scan costs 21 kernels per region. On the UDS maps the median-PSF scan returns $\lambda = 5.6 \times 10^{-4}$; a scan on one region alone returns 1.8×10^{-3} , so the representative pair matters. Cached kernel geojson files are reloaded without recording the method that produced them, so a map built with another method has to be rebuilt with `overwrite=True`.

3 Verification

Two profile families, both on a $0.04''$ grid of 401×401 pixels. The STPSF model grids in `data/PSF` for F444W and F770W, resampled with flux conserved and recentred, give 0.9302 within the $4.089''$ F444W field and 0.9912 within the $8.097''$ F770W field for the planes used; across all planes the values span 0.9179 to 0.9809 (F444W, 25 planes) and 0.9659 to 1.0030 (F770W, 9 planes). Two Moffat pairs with high-band FWHM $0.145''$ and low-band $0.28''$ bracket them: $\beta = 1.1$ has fat wings and keeps only 0.75 of its flux inside $4''$, $\beta = 2.5$ is compact.

Stamps are even-sized and cut with the slices of a real `Template` cutout, kernels are built from the stamp arrays, and the convolution is `Template.convolve_cutout` with the model pasted back onto the parent grid, so the test runs the production code path. The source is a noise-free point source of unit flux and the data are the untruncated low-resolution profile. Recovered model centroids sit within 0.02 to 0.07 pixels of the truth, which is the asymmetry of the PSFs themselves.

pair	stamp	S_{hi}	S_{lo}	wiener		window	
				A/S_{lo}	A/S_{hi}	A/S_{lo}	A/S_{hi}
STPSF	1.5''	0.9494	0.8506	1.00140	0.89717	1.03327	0.92573
	2''	0.9667	0.8789	1.00148	0.91049	1.03109	0.93740
	3''	0.9896	0.9158	1.00155	0.92693	1.02452	0.94819
	4''	0.9999	0.9410	1.00160	0.94264	1.02530	0.96494
	8''	1.0000	0.9996	1.00160	1.00117	1.02335	1.02291
Moffat 1.1	1.5''	0.6211	0.5286	1.00427	0.85471	1.01652	0.86513
	2''	0.6747	0.5944	1.00235	0.88306	1.01283	0.89229
	3''	0.7513	0.6893	1.00076	0.91820	1.00906	0.92582
	4''	0.7982	0.7477	1.00037	0.93713	1.00878	0.94500
	8''	0.9059	0.8823	1.00007	0.97399	1.00784	0.98156
Moffat 2.5	1.5''	0.9965	0.9772	1.00005	0.98065	1.01799	0.99824
	2''	0.9985	0.9895	1.00002	0.99101	1.01620	1.00704
	3''	0.9996	0.9969	1.00001	0.99733	1.01267	1.00996
	4''	0.9998	0.9986	1.00001	0.99882	1.01291	1.01171
	8''	1.0000	0.9998	1.00000	0.99987	1.01186	1.01173

Table 2: Recovery of an injected point source of unit flux. S_{hi} and S_{lo} are enclosed fractions of the model grid within the square stamp; exact recovery is $A/S_{\text{lo}} = 1$. With the Wiener kernel the JWST column is flat to 2×10^{-4} while S_{lo} moves 15 percentage points, and the largest departure in the table is 0.43%. A/S_{hi} moves by 10, 12 and 1.9 percentage points over the same rows.

3.1 S_{lo} is the divisor

S_{hi} reaches unity at 4 and 8'' for the JWST pair because the F444W model grid is only 4.089'' across. The detection PSF model is the more truncated of the two, missing 7.0% of its flux against 0.9% for F770W. Enlarging `psf_size` cannot recover that; it needs NIRCcam grids at a larger field of view.

3.2 The $S_{\text{hi}}-S_{\text{lo}}$ mismatch does not matter

Holding the stamp and the high-resolution profile fixed and varying only the low-resolution wing content:

β_{lo}	S_{lo}	$S_{\text{hi}}/S_{\text{lo}}$	A/S_{lo} wiener	A/S_{lo} window
1.1	0.7477	1.3371	1.00010	1.00853
1.3	0.8689	1.1506	1.00003	1.00953
1.6	0.9573	1.0444	1.00001	1.01072
2.0	0.9910	1.0089	1.00001	1.01188
3.0	0.9998	1.0001	1.00001	1.01366
6.0	1.0000	0.9998	1.00000	1.01592

Table 3: Mismatch at a fixed 4'' stamp, $S_{\text{hi}} = 0.99981$ throughout. Recovery is exact to 10^{-4} at every mismatch, and the small residual shrinks as the mismatch grows. The window error runs the other way, largest where the two profiles are most alike and the true kernel is closest to a delta function.

3.3 Weight maps and truncation

Repeating the JWST 4'' case with four weight maps gives $A/S_{\text{lo}} = 1.00160$ (uniform), 1.00130 (radial ramp over a factor 100), 1.00150 (uniform random 0.1 to 10) and 1.00214 (a zero-weight disk of radius 10 pixels over the core): a spread of 0.08% against a correction of 6.3%, as equation

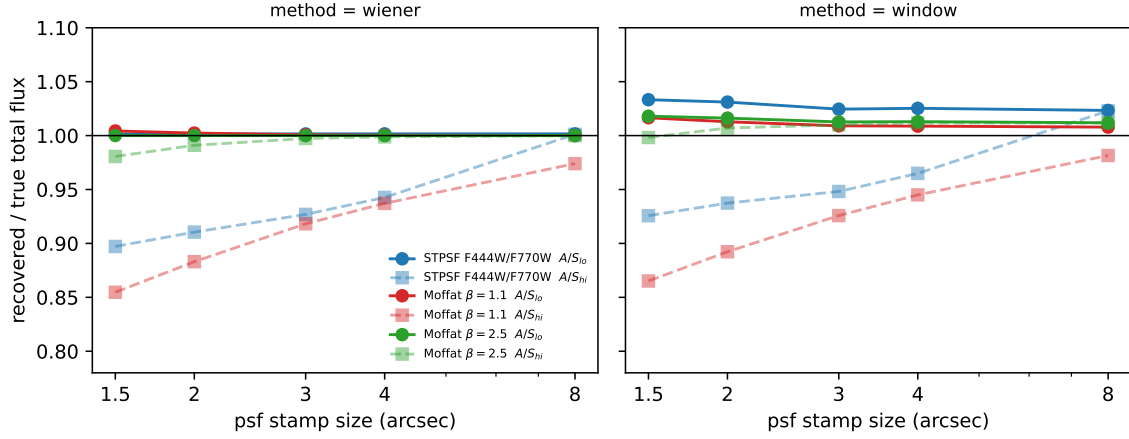


Figure 4: Recovered over true flux against stamp size for the two candidate divisors. Circles use S_{lo} , squares S_{hi} . Only the S_{lo} curves are flat. The offset of the right panel is the window’s error.

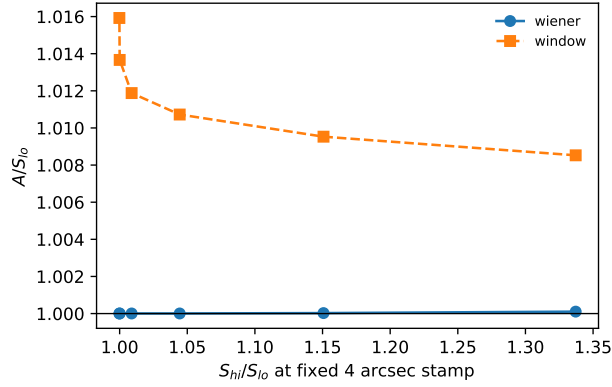


Figure 5: Recovery against mismatch at fixed stamp size.

3 requires. Template truncation behaves differently, the equivalent correction moving by 16% under the same masked core, which is why the two are kept apart.

Truncating the PSFs does change the kernel, so `psf_size` is a physical choice rather than a bookkeeping one. Largest kernel difference relative to the 8'' kernel, in units of its peak, after normalising both:

stamp	STPSF	Moffat 1.1	Moffat 2.5
1.5''	0.116	0.094	0.0169
2''	0.100	0.071	0.0086
3''	0.080	0.043	0.0034
4''	0.062	0.032	0.0026

Table 4: Renormalising a truncated PSF asserts that all its flux lies inside the stamp, and the kernel changes to reproduce that assertion. The difference is concentrated in the core, not the wings: for the JWST pair at 1.5'' it is 0.1164 of peak inside r_{lim} and 0.0053 outside.

Since $1/S_{lo}$ extrapolates the PSF model beyond the stamp, a larger stamp asks less of the model: at $S_{lo} = 0.94$ it supplies 6%, at 0.9998 almost nothing. Table 2 shows recovery is stable as the stamp grows, and the F770W grid extends to 8''. Below 1.5'' the deconvolution is poorly conditioned; 2'' is the smallest size at which all three profile families behave.

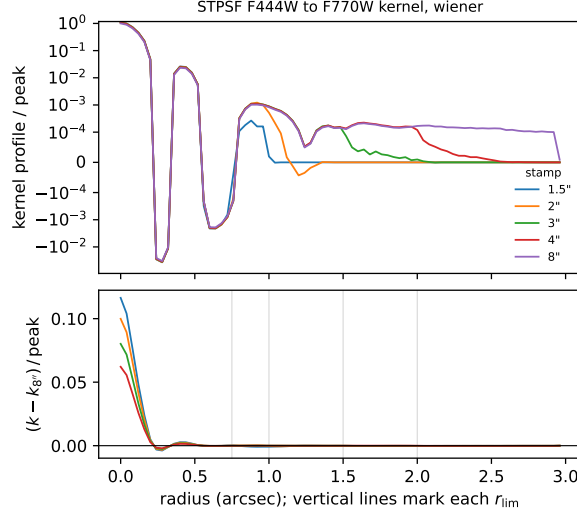


Figure 6: Top: azimuthally averaged kernel profile for the JWST pair at each stamp size, normalised to unit sum and divided by the peak, on a symmetric-log scale so the negative lobes show. Bottom: the difference from the 8'' kernel in units of its peak, which is the quantity in Table 4. The change sits inside $r \lesssim 0.25''$, in the kernel core, and reaches 12% of peak for the smallest stamp. Truncation alters the kernel, it does not merely clip it.

4 The matching kernel sets the residual

The window method biases the JWST flux scale by 1.0333, 1.0311, 1.0245, 1.0253, 1.0234 from 1.5 to 8'', and by 1.0180 to 1.0119 for the compact Moffat pair. The Wiener kernel leaves 1.0014 to 1.0016 on the JWST pair, flat to 2×10^{-4} , and is exact to five decimals on the compact Moffat pair.

The window bias has a closed form. `photutils` builds $\hat{k} = (\hat{p}_{\text{lo}}/\hat{p}_{\text{hi}})W$, so the matched model is $M = \mathcal{F}^{-1}[\hat{p}_{\text{lo}}W]$: the low-resolution shape passed through a low-pass filter, fitted against data that are not filtered. By Parseval, with uniform weights,

$$\frac{A}{S_{\text{lo}}} = \frac{\sum_f W(f) |\hat{p}_{\text{lo}}(f)|^2}{\sum_f W(f)^2 |\hat{p}_{\text{lo}}(f)|^2} \geq 1, \quad (4)$$

the inequality from $0 \leq W \leq 1$. The ratio depends on the window and the low-resolution power spectrum and on nothing else, which is why it does not move with the stamp: 2.2% for the JWST pair, 1.1% for the compact Moffat and 0.7% for the fat-winged one, larger where more PSF power lies above the window rolloff.

Equation 4 accounts for the whole offset at large stamps and most of it at small ones. The excess grows from 0.12% at 8'' to 1.1% at 1.5'' and comes from building the kernel on a grid barely larger than the object it has to represent.

Any method whose matched model reproduces p_{lo} without a deliberate low-pass leaves $A/S_{\text{lo}} = 1$, which is what the regularised methods do when the regularisation is small enough.

5 Scope

The measurements are for point sources. A resolved source needs the same S_{lo} and one further factor, the fraction of the source captured by the template's own support, $\text{EE}_{\text{hi}}(\Omega_t)/S_{\text{hi}}$. That factor is not a clean scalar: it depends on the source profile at fixed enclosed fraction, on the weight map and on the band, and in a blend it is not multiplicative at all. It is the reason templates are extended rather than corrected, and it is treated separately.

pair	stamp	predicted	measured	difference
STPSF	1.5''	1.02216	1.03327	0.0111
	3''	1.02215	1.02452	0.0024
	8''	1.02215	1.02335	0.0012
Moffat 1.1	1.5''	1.00771	1.01652	0.0088
	8''	1.00744	1.00784	0.0004
Moffat 2.5	1.5''	1.01126	1.01799	0.0067
	8''	1.01126	1.01186	0.0006

Table 5: Equation 4 evaluated from the window and the truncated low-resolution shape alone, with no kernel built, against the measured recovery. Full table in `ee_report/ee_window_cause.csv`.

The STPSF grids are resampled onto a common grid rather than drizzled, so the drizzle step is not exercised here. All tests are noise-free, so the regularisation optimum found on them does not transfer to real data and the scan must be run on the products in use. Totals are relative to the model grid, whose own absolute encircled energy is quoted in Section 3 and multiplies the numbers reported. Stamps are square, matching `psf_size`; a circular apodisation would move S_{lo} by the corner flux.

6 Summary

1. For a point source $A = f S_{lo}$. One scalar, the absolute encircled energy of the low-resolution PSF stamp, converts amplitude to total flux. The high-resolution truncation cancels through the kernel. It is measured per PSF region, recorded per template as `ee_psf_lo`, and applied once.
2. `ee_tmpl`, the fraction of the template surviving neighbour blanking, is recorded and not applied. The amplitude scales with the leverage of the blanked pixels, not with their flux, and those pixels lie in the faint wings; dividing by `ee_tmpl` over-corrects in every configuration tested.
3. The correction is independent of the weight map to 0.08%, and of the S_{hi}/S_{lo} mismatch to 10^{-4} over the range 1.34 to 1.00.
4. Normalising the kernel inputs rescales the kernel and leaves its shape unchanged to 10^{-15} . Truncating them changes the shape by 0.3 to 12% of peak, so `psf_size` is a physical choice.
5. The residual is set by the matching kernel: 2.3 to 3.3% for the Fourier window, of which a stamp-independent 2.2% follows from equation 4, against 0.16% for the regularised Wiener kernel.
6. The F444W model grid spans 4.089'' against F770W's 8.097'', so the detection PSF is the more truncated model and misses 7% of its flux. That needs new grids, not a larger `psf_size`.

Code and data: `scratch/wren/ee_report/run_ee_tests.py`, `test_ee_tmpl.py`, `fig_flow.py` and the CSV files alongside them.